

A closer look at some of today's mobile power sources and a peek at what's just around the corner

So here you are, the snazzy bigshot—PDA in one hand, laptop in another, when all of a sudden, the battery dies. Blank, black screens both—who you gonna call? Owners of such gadgets find themselves becoming less mobile than ever, and are always on the lookout for some place to plug-in and charge up.

The kind of battery that is required to run these devices all day, whilst utilising all their features, can only be found under the bonnet of a car. Unless we're willing to lug a monster of a battery around for our mobile devices, we're in desperate need of better power solutions.

Power storage technologies have

advanced quickly over the last few years. But these advances have been outpaced by the requirements of newer, more feature-packed portable devices. All batteries used today rely on chemical reactions to produce current. The only difference is the chemicals used, and their construction. Engineers constantly struggle to extract the most out of batteries—tirelessly trying new materials and constructions to squeeze out maximum performance.

Batteries can be classified as primary or secondary cells—secondary cells can be recharged, whilst primary cells cannot. Most power hungry portable devices run on secondary cells.

The ideal battery should offer very high energy densities, have a long service life, and be really light and tiny. Good batteries today, have only some of these qualities. Hence we always settle for a compromise, with the choice of battery being determined by the device it is made for. A mobile phone would need a tiny battery, whereas a UPS system would need the maximum possible power, with less stress laid on its weight.

The most common battery types are: **Lead acid:** The lead acid battery has been around for more than a hundred years. These are bulky, and only suitable for applications where weight is not a factor—car batteries are the best example. This type of battery is capable of delivering high power useful for starting your car, but it has a relatively low energy density and high self-discharge. This is why your car doesn't start after your long vacation. Lead acid is the most economical battery for larger power applications, and is the preferred choice for hospital equipment, wheelchairs, emergency lighting and UPS systems.

Nickel cadmium (NiCd): NiCd batteries have been around since the 1950s. They have matured over time, and continue to deliver good performance for situations where long service life is most important. These batteries perform well past a thousand cycles, but require proper maintenance to do so. The problem with NiCd batteries is their low energy density and the memory effect. NiCds also remain popular because they are cheap, but contain toxic chemicals that

